Nathan Amouroux, Ph.D.

8 rue du bicentenaire, 74960, Annecy, France
nathanamouroux@gmail.com
+33 6 67 60 38 42
github.com/NathanAmouroux
Born: 07 June 2000 (Age: 25)

Education

10/2023 – 09/2026 **Ph.D in Subatomic and Astroparticle Physics**, Laboratoire d'annecy de Physique des Particules (LAPP), Annecy, France.

2020 – 2023 **Magister degree in Physics**, Université Grenoble Alpes (UGA), Grenoble, France.

2022 – 2023 **Master degree in Subatomic Physics and Cosmology**, Université Grenoble Alpes (UGA), Grenoble, France.

2021 – 2022 **Master degree in Fundamental Research in Physics**, Université Grenoble Alpes (UGA), Grenoble, France.

2020-2021 **Bachelor degree in Physics**, Université Grenoble Alpes (UGA), Grenoble, France.

2018 – 2020 **Preparatory Classes for the Grandes Écoles (CPGE), Technology, Physics and Chemistry (TPC)**, Lycée Jean Mermoz, Montpellier, France.

Research Experience

10/2023 - 09/2026 **Ph.D. in Cosmology: Instrument Calibration and Foreground Mitigation for Precision Cosmology with the Vera C. Rubin Observatory**, *Laboratoire d'Annecy de Physique des Particules (LAPP), Annecy, France*

- Characterization of the Rubin telescope instrumental response with the Collimated Beam Projector (CBP).
- Development of the workflow and software for the analysis of the CBP data.
- Development and implementation of the foreground objects masking strategy for the cosmological analysis of the LSST data.

Advisor: Thibault Guillemin.

02/2023 - 06/2023 **Master 2 internship: Study of AMICO galaxy cluster detection algorithm performance for the selection of galaxy clusters in Rubin-LSST**, *Laboratoire d'Annecy de Physique des Particules (LAPP), Annecy, France*

- Evaluation of the performance of the AMICO galaxy cluster detection algorithm on simulated Rubin-LSST data.
- First comparisons of the performance of the AMICO algorithm with other galaxy cluster detection algorithms.

Advisor: Thibault Guillemin.

06/2022 - 07/2022 **Master 1 internship: Impact of the modelisation choice in galaxy clusters mass reconstruction with gravitational lensing**, *Laboratoire de Physique Subatomique et de Cosmologie (LPSC), Grenoble, France*

- Evaluation of the impact of the choice of the mass distribution model in galaxy clusters mass reconstruction with gravitational lensing.
- Comparison of the performance of different mass distribution models on simulated data.

Advisors: Céline Combet, Constantin Payerne.

Research Experience (continued)

06/2021 - 07/2021

Bachelor 3rd year internship: Micro-swimmers crowd dynamics, *Laboratoire Interdisciplinaire de Physique (LIPhy), Grenoble, France*

- Culture of algae (micro-swimmers) and creation of molds and masks for microfluidic devices.
- Study of the collective dynamics of micro-swimmers in a confined environment.
- Software development for the analysis of micro-swimmer trajectories.

Advisors: Salima Rafai, Eduardo Al Alam.

Collaboration Memberships

- LSST DESC member
- Vera Rubin Observatory Commissioning and calibration team member
- LSST France member

Research Publications

Journal Articles

1. C. Payerne et al., "Weak lensing mass-richness relation of redMaPPer clusters in LSST DESC DC2 simulations," *Astronomy and Astrophysics*, 2025.
2. R. Solomon et al., "Validation of the WaZP Cluster Finder on DC2 Simulations," 2025, Manuscript in preparation.
3. LSST Dark Energy Science Collaboration, "A Rubin View of Abell 360," 2025, Manuscript in preparation.

Conference Proceedings

1. N. Amouroux et al., *Early Telescope Throughput Results from the Collimated Beam Projector at the Vera C. Rubin Observatory*, **SPIE Astronomical Telescopes + Instrumentation**, Manuscript in preparation, 2026.
2. N. Amouroux, *Masking bright objects with LSST*, **Journées de Rencontres Jeunes Chercheurs**, 2024.
3. C. Payerne et al., *Multipole analysis of cluster weak lensing shear in The Three Hundred project*, **Observing the Universe at millimetre wavelengths**, 2023.

Technical Notes

1. E. K. Urbach et al., *Collimated Beam Projector Installation and ComCam Testing*, Rubin Technical Note RTN-152, 2025. URL: `https://rtn-152.lsst.io/`
2. Rubin Observatory Team, *The Vera C. Rubin Observatory Data Preview 1*, Rubin Technical Note RTN-095, 2025. URL: `https://rtn-095.lsst.io/`

Oral Presentations

2026 **SPIE Astronomical Telescopes + Instrumentation** *Copenhagen, Denmark* (upcoming)
- Poster: Early Telescope Throughput Results from the Collimated Beam Projector at the Vera C. Rubin Observatory

LSST France, *Annecy, France*
- Talk: LSSTCam filter throughput measurements with the CBP

2025 **LSST France**, *Grenoble, France*
- Talk: CCD QE measurements with the Collimated Beam Projector

LSST France, *Orsay, France*
- Talk: Collimated Beam Projector - Updates

2024 **LSST France**, *Marseille, France*
- Talk: DC2 bright stars masks

Journées Rencontres Jeunes Chercheurs, *Saint-Jacut-de-la-Mer, France*
- Talk: Masking bright objects with LSST

2023 **LSST France**, *Grenoble, France*
- Talk: Performances of the AMICO galaxy cluster algorithm on cosmoDC2 simulation

Cosmology Schools

2025 **Rodolphe Clédassou Summer School (past Euclid) Year 2 - Observational courses**, *Les Karellis*, The science of future major cosmological surveys

2024 **Rodolphe Clédassou Summer School (past Euclid) Year 1**, *Hendaye*, The science of future major cosmological surveys

Teaching

2024-2025 **Teaching assisting** Tutorials in Geometrical Optics for undergraduate students at Université Savoie Mont Blanc (USMB) $\sim 27h$

Teaching assisting Practical sessions in computational mathematics for undergraduate students in Electrical and Computer Engineering at Annecy IUT $\sim 60h$

2020-2023 **Tutorat** Support sessions for high school and first-year university students in sciences. $\sim 200h$

Skills

Languages	French (native), English (fluent), Spanish (intermediate).
Programming languages	Python, LaTeX, Shell / Bash
Python packages	Astropy, Batoid, emcee, healpy, ImSim, NumPy, Pandas, SciPy
Software	DS9, Git, Jupyter, LSST Science Pipelines, TOPCAT
Misc.	Academic research, teaching and mentoring, scientific communication, numerical simulations, cosmological data analysis, teamwork in large collaborations.

Outreach

2024–2026 Project leader for the outreach comic book *"L'Odyssée Cosmique de Pénélope"* about Ph.D. student work and focusing on 6 LAPP/LAPTh thesis (cosmology and astroparticles), *Annecy, France*.

2025 Speaker and discussion facilitator at the *European Researchers' Night*, *Chambéry, France*.

Facilitator for the *Rubin First Look* public event, *LPSC Grenoble, France*.

Participant in the *Rubin First Look* event, *LAPP Annecy, France*.

Speaker at the *Science Brewery* public discussions, *Annecy, France*.

2024 Presentation of the Vera C. Rubin Observatory at the *Forum of Former Students*, *Montpellier, France*.

Animator on the CNRS/IN2P3 – Vera C. Rubin Observatory booth during the *Journées de Rencontres du Ciel et de l'Espace*, *Paris, France*.

Contributor to the *Les Infinis à l'École* educational project, *Annecy, France*.

2023–2025 Animator of several *Mercredis du LAPP* outreach events, *Annecy, France*.

2023 Presentation of the MOND theory at *Anima Science*, *LAPP Annecy, France*.

Workshop organization

2025 Project leader for the organization of the Enigmass+ doctorants and post-doctorants workshop at IPAG and LAPP, Grenoble - Annecy, France.

Hobbies

- **Sports**: Running, Road bike, Swimming, Hiking
- **Video Games**
- **Mecanics / DIY**
- **Astronomy** (beginner)

References

- **Parker Fagrelius**, NoirLab, parker.fagrelius@noirlab.edu
- **Thibault Guillemin**, LAPP (Annecy), guillemin@lapp.in2p3.fr